Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In pure silicon at $300\ \mathrm{K}$ the electron and hole concentration is each equal to $1.5 \times 10^{16}\ \mathrm{m}^{-3}$. When doped with indium, the hole concentration increases to $4.5 \times 10^{22}\ \mathrm{m}^{-3}$. What is the electron concentration in doped silicon?

(A) $3 \times 10^{9}\ \mathrm{m}^{-3}$

(B) $4 \times 10^{9}\ \mathrm{m}^{-3}$

(C) $5 \times 10^{9}\ \mathrm{m}^{-3}$

(D) $6 \times 10^{9}\ \mathrm{m}^{-3}$

Q.2

A block of pure silicon at $300\ \mathrm{K}$ has a length of $10\ \mathrm{cm}$ and an area of $1.0 \times 10^{-4}\ \mathrm{m}^{2}$. If a battery of emf $2\ \mathrm{V}$ is connected across it, what is the electron current? The mobility of electrons is $0.14\ \mathrm{m}^{2}\ \mathrm{V}^{-1}\ \mathrm{s}^{-1}$ and their number density is $1.5 \times 10^{16}\ \mathrm{m}^{-3}$.

(A) $6.72 \times 10^{-4}\ \mathrm{A}$

(B) $6.72 \times 10^{-5}\ \mathrm{A}$

(C) $6.72 \times 10^{-6}\ \mathrm{A}$

(D) $6.72 \times 10^{-7}\ \mathrm{A}$

Q.3

When a $p-n$ junction made from silicon or germanium is forward biased, energy is released at the junction due to recombination of electrons and holes. This energy lies in the

(A) visible region

(B) infra-red region

(C) ultraviolet region

(D) gamma ray region

Q.4

What is the number density of donor atoms which must be added to a pure germanium semiconductor to produce an n-type semiconductor of conductivity $6.4 \Omega^{-1}\ \mathrm{cm}^{-1}$? The mobility of electrons in n-type germanium is $4 \times 10^{3}\ \mathrm{cm}^{2}\ \mathrm{V}^{-1}\ \mathrm{s}^{-1}$. Neglect the contribution of holes to conductivity.

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(A) $10^{16}\ \mathrm{m}^{-3}$

(B) $10^{18}\ \mathrm{m}^{-3}$

(C) $10^{20}\ \mathrm{m}^{-3}$

(D) $10^{22}\ \mathrm{m}^{-3}$

Q.5

The output from a full wave rectifier is

(A) an ac voltage

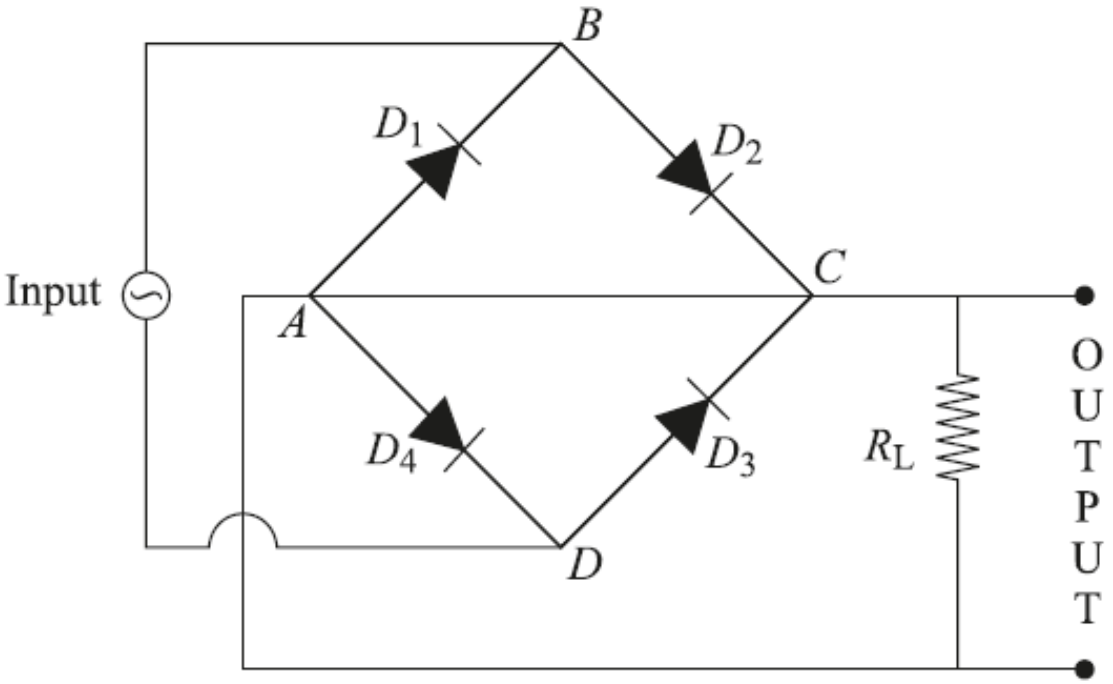
(B) a de voltage

(C) zero

(D) a pulsating unidirectional voltage

Q.6

The circuit shown in Fig. consists of four diodes D_1, D_2, D_3 and D_4. An alternating input voltage is applied across B and D and the output voltage is taken across a load resistor R_{L} connected between A and C. The output voltage is



(A) half wave rectified

(B) full wave rectified

(C) a d.c. voltage without ripple voltage

(D) zero

Q.7

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For a transistor, current gain $\alpha = 0.96$. It is used as an amplifier in a common base circuit with a load resistance of $4\text{k}\Omega$. If the dynamic resistance of the emitter-base junction is 48Ω, the voltage gain is

(A) 40

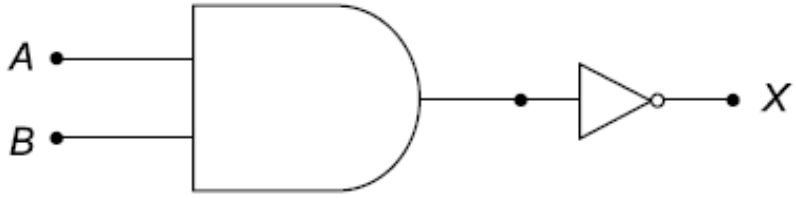
(B) 80

(C) 120

(D) 160

Q.8

What is the Boolean expression for the gate circuit shown in Fig.



(A) $\overline{A \cdot B} = X$

(B) $A + B = X$

(C) $A \cdot B = X$

(D) $\overline{A + B} = X$

Q.9

A block of pure silicon at $300\ \text{K}$ has a length of $10\ \text{cm}$ and an area of $1.0 \times 10^{-4}\ \text{m}^2$. Battery of emf 2V is connected across it. The mobility of electrons is $0.14\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}$ and their number density is $1.5 \times 10^{16}\ \text{m}^{-3}$. What is the hole current? The mobility of holes is $0.05\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}$

(A) $2.0 \times 10^{-7}\ \text{A}$

(B) $2.2 \times 10^{-7}\ \text{A}$

(C) $2.4 \times 10^{-7}\ \text{A}$

(D) $2.6 \times 10^{-7}\ \text{A}$

Q.10

In a transistor amplifier, the two ac current gains α and β as defined as $\alpha = \delta I_C / \delta I_E$ and $\beta = \delta I_C / \delta I_B$. The relation between α and β is

(A) $\beta = \frac{1+\alpha}{\alpha}$

(B) $\beta = \frac{1-\alpha}{\alpha}$

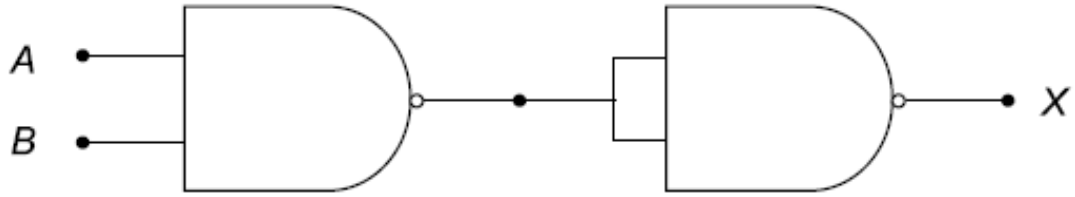
(C) $\beta = \frac{\alpha}{1-\alpha}$

(D) $\beta = \frac{\alpha}{1+\alpha}$

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Q.11

The combination of gates shown in Fig. yields



(A) AND gate

(B) NAND gate

(C) NOR gate

(D) NOT gate

Q.12

The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than $2480\ \text{nm}$ in incident on it. The band gap (in eV) for the semiconductor is

(A) 0.9

(B) 0.7

(C) 0.5

(D) 1.1

Q.13

In a transistor, a change of $8.0\ \text{mA}$ in the emitter current produces a change of $7.8\ \text{mA}$ in the collector current. What change in the base current is necessary to produce the same change in the collector current?

(A) $50\mu\text{A}$

(B) $100\mu\text{A}$

(C) $150\mu\text{A}$

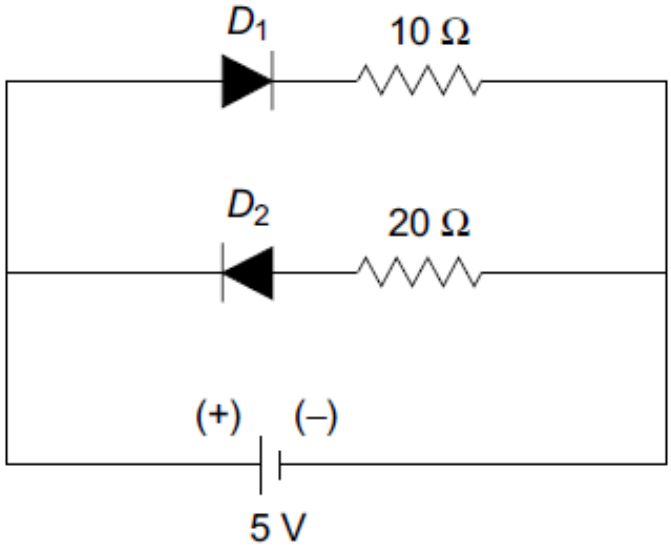
(D) $200\mu\text{A}$

Q.14

Two ideal junction diodes D_1 and D_2 are connected to a battery as shown in Fig. The current supplied by the

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battery is



(A) $0.25\ \mathrm{A}$

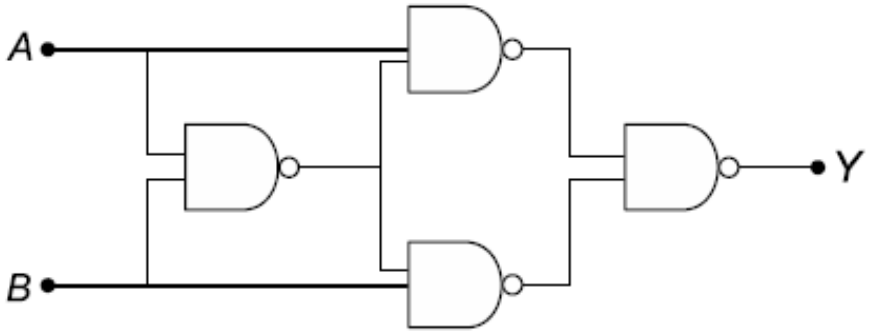
(B) $0.5\ \mathrm{A}$

(C) $0.75\ \mathrm{A}$

(D) zero

Q.15

The truth table for system of four NAND gates as shown in Fig. is



(A) Misplaced \hline

(B) Misplaced \hline

(C) Misplaced \hline

(D) Misplaced \hline

Q.16

The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than $2480\ \mathrm{nm}$ is incident on it. The band gap energy (in eV) for the semiconductor is

(A) $1.1\mathrm{eV}$

(B) $2.5\mathrm{eV}$

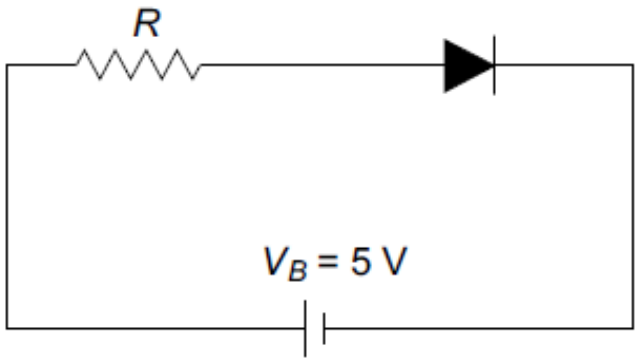
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(C) 0.5eV

(D) 0.7eV

Q.17

The knee point of the junction diode used in the circuit shown in Fig. is $0.5\ \text{V}$. A minimum current of $1.5\ \text{mA}$ is required for the diode to be above the knee point. The maximum value of R so that the voltage across the diode is above the knee point is (assume that, above the knee point, the voltage across the diode is independent of the current).



(A) 3333Ω

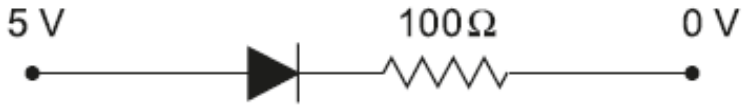
(B) 3000Ω

(C) 3667Ω

(D) 7500Ω

Q.18

A junction diode has a resistance of 25Ω when forward biased and 2500Ω when reverse biased. The current in the arrangement shown in Fig. will be



(A) $\frac{1}{15}$ A

(B) $\frac{1}{25}$ A

(C) $\frac{1}{520}$ A

(D) $\frac{1}{480}$ A

Q.19

In a semiconductor, it is found that $\frac{4}{5}$ th of the total current is carried by electrons and the remaining $\frac{1}{5}$ th by the holes. If at this temperature, the drift speed of electrons is 2.5 times that of holes, the ratio of the number densities of electrons and holes is

(A) $\frac{8}{5}$

(B) $\frac{5}{8}$

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(C) $\frac{25}{8}$

(D) $\frac{1}{2}$

Q.20

In an $n-p-n$ transistor circuit, the collector current is $10\ \mathrm{mA}$. If 90% of the electrons emitted reach the collector,

(A) the emitter current will be nearly $9\ \mathrm{mA}$

(B) the emitter current will be nearly $11\ \mathrm{mA}$

(C) the base-current will be nearly $0.9\ \mathrm{mA}$

(D) the base current will be nearly $-1\ \mathrm{mA}$

Q.21

The following table
\begin{array}{cc|c} \ A & B & X \\ \ \hline 0 & 0 & 1 \\ \ 1 & 0 & 1 \\ \ 0 & 1 & 1 \\ \ 1 & 1 & 0 \\ \ \hline \ \end{array}
is a truth table for

(A) NAND gate

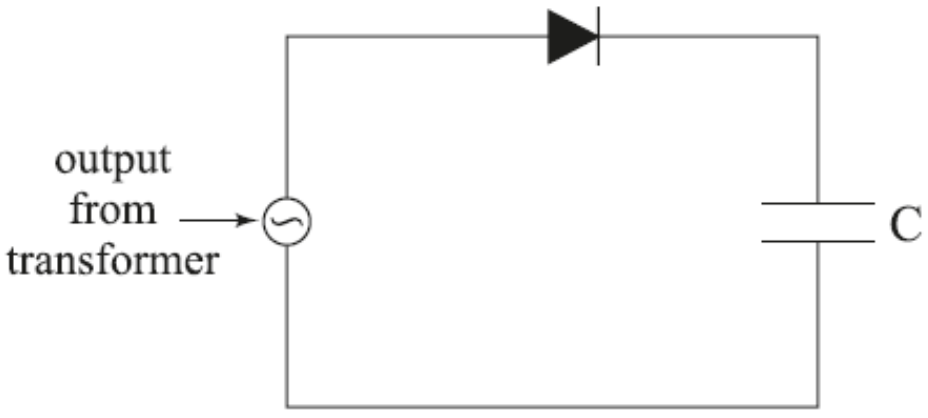
(B) NOR gate

(C) XOR gate

(D) AND gate

Q.22

An alternating voltage $V = 200\sin(\omega t)$ is connected across the primary of a step-down transformer of turns ratio $1:20$. The output of the transformer is rectified as connected in the half-wave rectifier circuit as shown in Fig. If $C = 5\mu\mathrm{F}$, the final charge on the capacitor plates is



(A) $20\mu\mathrm{C}$

(B) $50\mu\mathrm{C}$

(C) $50\sqrt{2}\mu\mathrm{C}$

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(D) $100\mu C$

Q.23

The truth tables of two logic gates P and Q are as follows.
\begin{array}{l} \text { Logic gate } P\\ \begin{array}{|c|c|c|} \hline A & B & X \\ \hline 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \\ \hline \end{array}\\ \text {Logic gate} Q \\ \begin{array}{|l|l|} \hline A & X \\ \hline 0 & 1 \\ 1 & 0 \\ \hline \end{array} \end{array}
When the output of gate P is connected to the input of gate Q, we get a new gate called

(A) OR gate

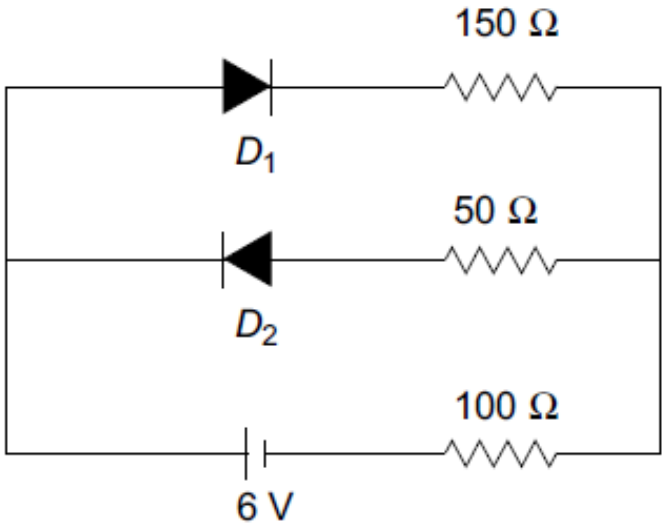
(B) AND gate

(C) NAND gate

(D) NOR gate

Q.24

The circuit shown in Fig. contains two diodes D_1 and D_2 each with a forward resistance of 50 ohms and with infinite backward resistance. If the battery voltage is $6\ \mathrm{V}$, the current through the $100\mathrm{ohm}$ resistance (in amperes) is



(A) zero

(B) 0.02

(C) 0.03

(D) 0.036

Q.25

In a transistor circuit, the collector current is $50\ \mathrm{mA}$ and the base current is $1\ \mathrm{mA}$. The current gain α is

(A) $\frac{51}{50}$

(B) $\frac{49}{50}$

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(C) $\frac{50}{51}$

(D) $\frac{49}{51}$

Answers & Solutions

Q.1 Answer:

$5 \times 10^{9}\ \mathrm{m}^{-3}$

Solution:

Given $n_i = 1.5 \times 10^{16}\ \mathrm{m}^{-3}$ **and** $n_h = 4.5 \times 10^{22}\ \mathrm{m}^{-3}$

Now, we know that

$$n_e n_h = n_i^2$$

$$\text{or} \quad n_e = \frac{n_i^2}{n_h} = \frac{\left(1.5 \times 10^{16}\right)^2}{4.5 \times 10^{22}} = 5.0 \times 10^{9}\ \mathrm{m}^{-3}$$

Q.2 Answer:

$6.72 \times 10^{-7}\ \mathrm{A}$

Solution:

Given $\mu_e = 0.14\ \mathrm{m}^2\ \mathrm{V}^{-1}\ \mathrm{s}^{-1}, n_e = 1.5 \times 10^{16}\ \mathrm{m}^{-3}, l = 10\ \mathrm{cm} = 0.1\ \mathrm{m}, A = 1.0 \times 10^{-4}\ \mathrm{m}^2$ **and** $V = 2$ **volts. The electric field in the block is**

$E = \frac{V}{l} = \frac{2}{0.1} = 20 \mathrm{Vm}^{-1}$

The drift speed of electrons is

$v_e = \mu_e E = 0.14 \times 20 = 2.8\ \mathrm{ms}^{-1}$

$\therefore$ **Electron current** I_e **is**

$$= n_{e^A e v_e}$$
$$= \left(1.5 \times 10^{16}\right) \times \left(1.0 \times 10^{-4}\right) \times \left(1.6 \times 10^{-19}\right) \times 2.8$$
$$= 6.72 \times 10^{-7}\ \mathrm{A}$$

Q.3 Answer:

infra-red region

Solution:

The correct choice is (b)

Q.4 Answer:

$10^{22}\ \mathrm{m}^{-3}$

Solution:

Given $\sigma = 6.4 \Omega^{-1}\ \mathrm{cm}^{-1} = 6.4 \times 10^2 \Omega^{-1}\ \mathrm{m}^{-1}$ **and**

$\mu_e = 4 \times 10^3\ \mathrm{cm}^2\ \mathrm{V}^{-1} \sigma^{-1} = 4 \times 10^{-1}\ \mathrm{m}^2\ \mathrm{V}^{-1} \sigma^{-1}$

Neglecting the contribution of holes, the expression for σ **is**

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$$\sigma = e n_e \mu_e$$

$$\text{or } n_e = \frac{\sigma}{e\mu_e} = \frac{6.4 \times 10^2}{(1.6 \times 10^{-19}) \times (4 \times 10^{-1})}$$

$$= 10^{22}\ \mathrm{m}^{-3}$$

Q.5 Answer:

a pulsating unidirectional voltage

Solution:

The correct choice is (d)

Q.6 Answer:

full wave rectified

Solution:

The given circuit is a full wave rectifier called a bridge rectifier. During the positive half cycle of the input, diodes D_2 **and** D_4 **are forward biased and hence they conduct, but diodes** D_1 **and** D_3 **are reverse biased and hence they do not conduct. During the next (negative) half cycle,** D_1 **and** D_3 **conduct but** D_2 **and** D_4 **do not. Since at a time two diodes conduct, the output voltage is higher in the case of a bridge rectifier than a full wave rectifier in which only two diodes are used instead of four. So a bridge rectifier is always preferred as a full wave rectifier. Thus the correct choice is (b).**

Q.7 Answer:

80

Solution:

Voltage gain $= \alpha \times \frac{R_L}{R_D} = 0.96 \times \frac{4000}{48} = 80$. **Hence the current choice is (b).**

Q.8 Answer:

$\overline{A \cdot B} = X$

Solution:

The circuit is a combination of AND and NOT gates. The output of the NAND gate is connected to the input of a NOT gate. The combined circuit is called a NAND gate. The Boolean expression for the NAND operation is

$\overline{A \cdot B} = X$

which means that the output X **is just the negation of the output of the AND operation. Hence the correct choice is (a).**

Q.9 Answer:

$2.4 \times 10^{-7}\ \mathrm{A}$

Solution:

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In a pure semiconductor $n_e = n_h = 1.5 \times 10^{16}\ \mathrm{m}^{-3}$ Given $\mu_h = 0.05\ \mathrm{m}^2\ \mathrm{V}^{-1}\ \mathrm{s}^{-1}$. The drift speed of holes is

$$
\begin{aligned}
v_h &= \mu_h E = 0.05 \times 20 = 1.0\ \mathrm{ms}^{-1} \\
\therefore \text{ Hole current } \quad I_h &= n_h A e v_h \\
&= \left(1.5 \times 10^{16}\right) \times \left(1.0 \times 10^{-4}\right) \\
&\quad \times \left(1.6 \times 10^{-19}\right) \times 1.0 \\
&= 2.4 \times 10^{-7}\ \mathrm{A}
\end{aligned}
$$

Q.10 Answer:

$\beta = \frac{\alpha}{1-\alpha}$

Solution:

We know that $I_E = I_C + I_B$ or $\delta I_E = \delta I_C + \delta I_B$. Dividing both sides by δI_C, we get

$$
\begin{aligned}
\frac{\delta I_E}{\delta I_C} &= 1 + \frac{\delta I_B}{\delta I_C} \\
\text{or} \quad \frac{1}{\alpha} &= 1 + \frac{1}{\beta} \\
\text{or} \quad \beta &= \frac{\alpha}{1-\alpha}
\end{aligned}
$$

Q.11 Answer:

AND gate

Solution:

Using a NOT gate after a NAND gate yields an AND gate. Hence the correct choice is (a).

Q.12 Answer:

0.5

Solution:

Wavelength of incident radiation is $\lambda = 2480\ \mathrm{nm} = 2480 \times 10^{-9}\ \mathrm{m}$. Energy of incident radiation is

$$
\begin{aligned}
E &= \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{2480 \times 10^{-9}} \\
&= 8.0 \times 10^{-20}\ \mathrm{J} \\
&= \frac{8.0 \times 10^{-20}}{1.6 \times 10^{-19}} \mathrm{eV} = 0.5 \mathrm{eV}
\end{aligned}
$$

Q.13 Answer:

$200 \mu\mathrm{A}$

Solution:

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$\delta I_E = 8.0 \times 10^{-3}$ A **and** $\delta I_C = 7.8 \times 10^{-3}$ A. **Now**

$\alpha = \frac{\delta I_C}{\delta I_E} = \frac{7.8 \times 10^{-3}}{8.0 \times 10^{-3}} = 0.975$

$\therefore \quad \beta = \frac{\alpha}{1-\alpha} = \frac{0.975}{1-0.975} = 39$

$\therefore \beta = \dfrac{\delta I_C}{\delta I_B}$. Therefore

$$\text{But } \delta I_B = \frac{\delta I_C}{\beta} = \frac{7.8 \times 10^{-3}}{39}$$
$$= 2 \times 10^{-4} \text{ A} = 200 \mu\text{A}$$

Q.14 Answer:

0.5 A

Solution:

Diode D_1 is forward biased. Therefore, effective resistance is 10Ω. Hence the current is

$I = \frac{5 \text{ V}}{10\Omega} = 0.5 \text{ A}$

Thus the correct choice is (b).

Q.15 Answer:

\begin{array}{|c|c|c|} \hline \mathbf{A} & \mathbf{B} & \mathbf{Y} \\ \hline 0 & 0 & 0 \\ \hline 0 & 1 & 1 \\ \hline 1 & 0 & 1 \\ \hline 1 & 1 & 0 \\ \hline \end{array}

Solution:

$$\text{Y} = (\overline{A \cdot \overline{A \cdot B}}) \cdot (\overline{B \cdot \overline{A \cdot B}})$$
$$= A \cdot (\bar{A} + \bar{B}) + B \cdot (\bar{A} + \bar{B})$$
$$= A \cdot \bar{A} + A.B + B \cdot \bar{A} + B \cdot \bar{B}$$
$$= \text{O} + A \cdot \bar{B} + \text{B} \cdot \bar{A} + \text{O}$$

So the correct truth table is (a).

Q.16 Answer:

0.5eV

Solution:

$$E_g = h\nu = \frac{hc}{\lambda}$$
$$= \frac{\left(6.63 \times 10^{-34}\right) \times \left(3 \times 10^{8}\right)}{2480 \times 10^{-9}}$$
$$= 0.81 \times 10^{-19} \text{ J}$$
$$= \frac{0.81 \times 10^{-19}}{1.6 \times 10^{-19}} \simeq 0.5\text{eV}$$

Q.17 Answer:

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3000Ω

Solution:

The maximum voltage drop across R must be $5-0.5=4.5$ V. Hence maximum value of R is

$R_{\max}=\frac{4.5\text{ V}}{1.5\text{ mA}}=\frac{4.5\text{ V}}{1.5\times10^{-3}\text{ A}}=3000\Omega$

Thus the correct choice is (b).

Q.18 Answer:

$\frac{1}{25}$ A

Solution:

The junction diode is forward biased. Therefore, the effective resistance $=100+25=125\Omega$. The current is

$I=\frac{5\text{ V}}{125\Omega}=\frac{1}{25}\text{ A}$

Hence the correct choice is (b).

Q.19 Answer:

$\frac{8}{5}$

Solution:

$n_e e v_e+n_h e v_h=I$. **Given**

$\frac{n_e e v_e}{n_h e v_h}=\frac{4I/5}{I/5}=4$

$\frac{n_e v_e}{n_h v_h}=4$ or $\frac{n_e}{n_h}=4\frac{v_h}{v_e}=\frac{4}{2.5}=\frac{8}{5}$

Hence the correct choice is (a).

Q.20 Answer:

the emitter current will be nearly 11 mA

Solution:

For a transistor, $I_E=I_B+I_C$

We are given that $I_C=\frac{90}{100}I_E=0.9I_E$

Since $I_C=10\text{ mA}$**, we have**

$$I_E=10\times\frac{10}{9}\text{ mA}=11.1\text{ mA}\approx11\text{ mA}$$

Also $I_B=I_E-I_C$

$$=(11-10)\text{mA}\approx1\text{ mA}$$

Hence the correct choice is (2).

Q.21 Answer:

NAND gate

Solution:

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The correct choice is (a).

Q.22 Answer:

$50\mu\text{C}$

Solution:

Peak value of output voltage from the transformer is

$V_0 = \frac{200}{20} = 10\text{ V}$

The junction diode conducts during the half cycle of the input when it is forward biased. During this half cycle, the capacitor is charged to the peak value of the supply voltage (which is 10 V). Hence the final charge on capacitor plates is

$Q = CV_0 = \left(5 \times 10^{-6}\right) \times 10 = 50\mu\text{C}$

So the correct choice is (b).

Q.23 Answer:

NAND gate

Solution:

Gate P is an AND gate and gate Q is a NOT gate. Hence the new gate formed is a NAND gate. So the correct choice is (c).

Q.24 Answer:

0.02

Solution:

In the given circuit only diode D_1 will allow the current to pass through as it is forward-biased. Hence, the current through the 100 ohm resistance is

$(\because \text{ resistance of } D_1 = 50\Omega)$

$$I = \frac{6\text{ V}}{(150+50+100)\Omega} = \frac{6}{300}\text{ A}$$
$$= 0.02\text{ A}$$

Q.25 Answer:

$\frac{50}{51}$

Solution:

Current gain $\beta = \frac{I_C}{I_B} = \frac{50}{1} = 50$. Therefore, current gain α is

$\alpha = \frac{\beta}{1+\beta} = \frac{50}{1+50} = \frac{50}{51}$

Hence the correct choice is (c).

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